Lab Assignment 9
(Machine Learning, 307)
Decision Tree and Random Forest

Section 1: Decision Tree

Consider the dataset "Social_Network_Ads.csv" to predict the purchase.
Dataset link: https://www.kaggle.com/datasets/akram24/social-network-ads

1) Load the data (csv file) and print first 10 rows.
2) Check missing values in variables.
3) Separate the features into X (features) and Y (outcome) and print the shape.
4) Split data into separate training and test set in (80,20) ratio.
5) Implement Decision Tree Classifier
6) Predict on the test data.
7) Print the classification report , F1 score, accuracy rate, and confusion matrix.
8) Visualise the tree.

Section 2: Random Forest

Consider the dataset "diabetes_dataset" to predict the diabetes.
Dataset link: https://www.kaggle.com/datasets/mathchi/diabetes-data-set

1) Load the data (csv file), read the dataset into the data frame 'df' and print the different statistical values and shape of data.
2) Check missing values in variables.
3) Separate the features into X and Y and print the shape.
4) Split data into separate training and test set in (70,30) ratio.
5) Implement Random Forest classifier with [n_estimators=100 and Criterion = Entropy].
6) Find important features and print them. [import 'feature_importances' library]
7) Built another model with random forest classifier but only with the selected Important features (found in step 6).
8) Print the accuracy score for both the model.
9) Print and compare the confusion matrix for both the model.